LIVELIHOOD DIVERSIFICATION IN THE CONTEXT OF CLIMATE CHANGE: EVIDENCE FROM FARM HOUSEHOLDS IN SOUTH-EAST TUNISIA

Fatma ARIBI[1,2*], Mondher FETOUI[1], Mongi SGHAIER[1]

[1]Institute of Arid Regions of Medenine, LESOR Laboratory "Economy and Rural Societies", Tunisia
[2]National Agronomic Institute of Tunis, Carthage University, Tunisia
*Corresponding author:fatmaribi@gmail.com

Abstract

Livelihood diversification reduces farm household vulnerability to the adverse effect of climate change. This study aims to analyze livelihood diversification of the farm households in southeast Tunisia. The research used a multistage random sampling technique to collect data from 50 farm households. Data were gathered through structured questionnaires and were described by means of descriptive statistics. The Simpson diversity index (SDI) was used to assess the level of livelihood diversification of households. Furthermore, with reference to the sustainable livelihood framework, the study showed the difference of capital endowments between livelihood groups. It also highlighted the determinants of livelihood diversification using a logistic regression model. The results showed that households with high levels of SDI (SDI > 0.5) were likely to have more sources of income. Furthermore, there was a significant difference ($p < 0.05$) in capital endowments between households that diversified their livelihoods and those that did not. Households with high SDI are more endowed with human, social and financial capital. Results of the logistic model highlighted that level of education, farm size, herd size, non-farm income and distance to the main plot are the key determinants of livelihood diversification.

Keywords:*Livelihood diversification, households, Simpson index, Sustainable livelihood Framework, Logistic regression.*

Introduction

Uncertainties and risks induced by climate change effects have been increased especially in the agricultural sector. Therefore, enhancing the adaptive capacity and the resilience of farm households are needed to deal with these threats. Diversification of livelihood strategies is a key solution to reduce the adverse impacts of climate change (Saha and Bahal, 2015). The concept of livelihood strategies is generally recognized as the combination of choices and activities performed by households in order to attain their livelihood goals (Khatun and Roy, 2012). It includes farm activities as well as non-farm activities. Previous studies showed that the impact of risk in agriculture and the pressure on households' farm income has stimulated the search of alternate revenue sources. Further, it has been reported that in low-income countries, farming activities hardly grant sufficient incomes that meet the needs of households (Fabusoro et al., 2010). Due to this fact, most farm households depend on a diversified portfolio of income sources to spread risks and to stabilize their revenues (Ellis, 1999).

To understand the factors affecting the livelihood strategies, several scholars have used the Sustainable Livelihood Approach (SLA), which was developed by the UK Department for International Development (DFID) (Chambers, 1987; Carney, 2003; Samsudin and Kamaruddin, 2013). The SLA takes into account a complex range of assets on which households depend for their livelihoods (Karl, 2002). This range includes tangible assets (money, land, agricultural equipment) and intangible assets (skills, social networks) (Xu et al., 2015). There are different ways to conceptualize these assets, but the most common is in the form of a pentagon representing five types of capitals: human, financial, physical, social and

natural capital (Carney, 2003). The SLA forms the conceptual and methodological basis for the "livelihoods" analysis that will be carried out in this paper.

Several international studies have been conducted to analyze the households' livelihood diversification (Fabusoro et al., 2010; Saha and Bahal, 2015; Ahmed et al., 2018; Gecho, 2017). For instance, Babatunde and Qaim, 2009 have assessed the households' livelihoods diversification in rural Nigeria. They employed the Herfindhal index to assess the livelihood diversification and the Tobit model to identify the determinants of diversification. They found that the male gender, the education, the productive assets, the pipe borne water and the access to credit are the major drivers of livelihood diversification. Likewise, another study conducted in West Bengal has analyzed the determinants of livelihood diversification. It showed that the educational level, the household experience, the training, the access to credit, the social status and the agro-climatic condition are the major factors affecting the livelihoods diversification (Khatun and Roy, 2012).

However, in Tunisia, there is a current lack of studies investigating the households' livelihoods diversification. This research tries to fill this gap through a case study of households from Medenine province, which belongs to South-East Tunisia. The purpose of this paper is to assess the level of diversification of these farm households and to analyze its main determinants.

Materials and Methods

The study was carried out in the Oum Zessar watershed, which is located in the northwest of Medenine province. It covers an area of 350 Km^2. It is spread over a territory of 11 imadas belonging to three districts: Beni khedache, North Medenine and Sidi Makhlouf. A multistage random sampling technique was employed. In the first stage, the three districts were selected to cover the entire watershed. In the second stage, out of the 11 imadas of the watershed, nine were selected randomly. In the third stage, households were selected randomly from each imada. Overall, 50 farm households were selected for this study. Data were gathered using a survey questionnaire with farm households.

To evaluate the level of households' livelihood diversification, this research used the Simpson Diversity Index (SDI), which was established by Edward H. Simpson in 1949 (Ahmed et *al.*, 2018). The SDI calculates both the number of livelihoods practiced by a household and the proportional contribution of each livelihood. We considered three types of livelihood strategies: crop-oriented, livestock oriented and off-farm-oriented. The SDI is calculated using the following equation:

$$\mathbf{SDI = 1 - \sum_{i=1}^{N} P_i^2} \qquad (1)$$

Where : N represents the number of income sources and P_i is the income proportion of the i-th income source. The value of the SDI is between 0 and 1. Households with less diversified livelihoods are associated with the littlest values of SDI. Those with a highly diversified income have a high SDI value. The SDI was calculated for each household and was used to group sample households into two groups according to their level of diversification. The level of livelihood diversification was interpreted using the SDI values: if SDI < 0.5, the household's livelihoods are lowly diversified; if SDI > 0.5 the household's livelihoods are highly diversified. Then, using the Sustainable Livelihood Approach (SLA), we compared the capital endowment between the two households' groups. Indicators used to quantify each capital (Table 1) were selected based on a literature review and on the availability of data in our database. Overall 17 indicators were chosen to apply the SLA framework. It was crucial to standardize each indicator because of the heterogeneity of units obtained from the raw data, the following formula was used:

$$\mathbf{Index\ I_h = (I_h - I_{min}) / (I_{max} - I_{min})} \qquad (2)$$

Downloaded from https://cabidigitallibrary.org by 196.203.128.101, on 04/23/26.
Subject to the CABI Digital Library Terms & Conditions, available at https://cabidigitallibrary.org/terms-and-conditions

Where I_h is the original variable of household h, I_{min} and I_{max} are respectively the lowest and highest value of variable among all households.
In this paper, the decision to highly diversify livelihoods by households or not is considered a dichotomous variable, hence the use of a logistic regression model (McFadden, 1973).
For this dichotomous model, a logistic regression analysis using maximum likelihood was performed to test the effect of several explanatory factors on the probability of "livelihood diversification by households". If multicollinearity problem exists in the model, it is difficult to determine the effects of the explanatory variables on the dependent variable. Therefore, the multicollinearity problem between explanatory variables was tested before running the logistic model. Once we find two correlated variables, one of them should be eliminated to solve this problem (Ahmed et al., 2018). Finally, as shown in eqation (3), seven variables were maintained to run the logistic model. The obtained model is as below:

$$\mathbf{LDI = \beta_0 + \beta_1\ Education + \beta_2\ Farm\ size + \beta_3\ Herd\ size + \beta_4\ Distance\ to\ the\ main\ plot + \beta_5\ Experience\ in\ agriculture + \beta_6\ Non\text{-}farm\text{-}income + \beta_7\ Distance\ to\ the\ market + \mu_i} \quad (3)$$

Where, LDI: Livelihood diversification index, β_0 = Intercept, μ_i = Error term

Results and Discussion

Study's results showed that all households are diversifying their livelihoods, while, the level of diversification differs among households. The average value of SDI for the whole sample was 0.41 with a standard deviation equal to 0.45. This indicates a great variability in the levels of diversification between households. This led us to classify the households into two groups using the binary logistic regression (0 for households with low level of diversification and 1 otherwise).

Table 1. Difference in capital endowment between the two groups of households

Livelihood Capital	Indicators	Group 1 (SDI<0.5)	Group 2 (SDI>0.5)
Human capital	Age	0.700	0.333
	Household size	0.448	0,614
	Dependency ratio	0.185	0.273
	Number of years of education of head of household	0.434	0.698
	Trainings	0.306	0.607
	Number of years of experience in agriculture	0.534	0.221
Physical capital	Value of agricultural equipment	0.308	0.002
	distance from the main plot	0.010	0.179
	Herd size	0.320	0.160
Natural capital	Farm size	0.308	0.154
	Irrigated area	0.149	0.002
	Number of olive tree	0.100	0.274
	Crop diversity index	0.412	0.189
Social capital	Membership in any organization	0.136	0.536
	Distance to the market	0.130	0.427
Financial Capital	Non-farm income	0.339	0.658
	Subsidies	0.089	0.000

Downloaded from https://cabidigitallibrary.org by 196.203.128.101, on 04/23/26.
Subject to the CABI Digital Library Terms & Conditions, available at https://cabidigitallibrary.org/terms-and-conditions

Group 1 encompass 54% of the surveyed households that have a low level of diversification (SDI<0.5). Nevertheless, the other 46% have high SDI values. Table 1 and figure 1 compares the capital endowment between the two groups. There was a difference in the amount of livelihood capital between the two groups. Group 1 possess more abundant physical (0.212) and natural capital (0.242). Group 2 owns more abundant human (0.457), financial (0.328) and social capital (0.48). These differences were statistically significant (Figure 1).

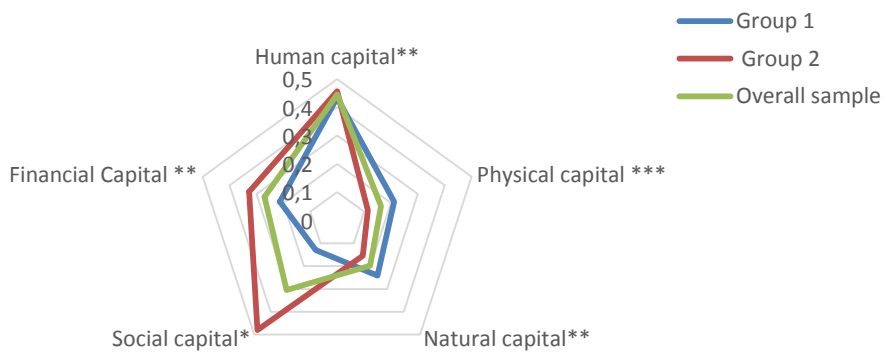

Figure 1. Difference in capital endowment between the two groups of households (*1% significance, ** 5 % significance, *** 10% significance)

Table 2 presents the results of the logit model estimation (Y) which explains the probability that a household will highly diversify its livelihoods. The model is statistically valid. Indeed, the validity of the estimated model is analyzed through the Wald $\chi^2(5) = 10.48$ statistic, which tests the equality to zero of all the model coefficients. On the other hand, he showed that the coefficients retained in the model were significantly different from zero at the 5 % threshold. Consequently, the hypothesis of nullity of the coefficients is rejected. The Wald χ^2 tested also the link between each of explanatory variables and the household's livelihoods diversification.

Using R^2 McFadden, good adequacy of the models was demonstrated. In fact, R2 has a satisfactory high value R2 =0.66, which prove the model performance.

Table 2. Predictability of the Logit model

Logit (LDI)	Coefficients	Pr > Chi²
Education	2.999	0.005*
Distance to the main plot	0.968	0.038**
Experience in agriculture	-0.521	0.205
Farm size	-2.364	0.014**
Herd size	1.126	0.089***
Distance to the market	0.756	0.182
Non-farm-income	0.928	0.092***
Wald χ^2	10.481	
R^2(McFadden)	0.662	
AIC	69.41	

*1% significance, ** 5 % significance, *** 10% significance

Then, we tested the independence of the explanatory variables with each other (Table 3). This led to a reduction in the number of variables and allowed us to avoid multi-collinearity. To test the existence of multi-collinearity, the explanatory variables were verified using a

Downloaded from https://cabidigitallibrary.org by 196.203.128.101, on 04/23/26.
Subject to the CABI Digital Library Terms & Conditions, available at https://cabidigitallibrary.org/terms-and-conditions

variance inflation factor (VIF) which makes it possible to detect the multi-collinearity of the explanatory variables (Asfaw et al., 2017).
In our model, there is no multi-collinearity between the variables (Table 3). In fact, a variance inflation factor (VIF) and its inverse (1 / VIF) are used to verify the explanatory variables.
(1 / VIF) is greater than 0.1. Therefore, it can be concluded that the model variables explain significantly the household's livelihoods diversification.
Results analysis of the Logit econometric model (Y) presented in Table 2 shows that the significant variables were: level of education (1% threshold); land size and the distance to the main plot (5% threshold) and herd size and the non-farm income (10% threshold).

Table 3. Variance Inflation Factor (VIF)

Variables	VIF	1/VIF
Education	3,666	0,27275655
Distance to the main plot	1,281	0,78047636
Experience in agriculture	1,802	0,55486122
Land size	1,552	0,6441241
Herd size	1,190	0,84043628
Distance to the market	1,891	0,52881396
Non-farm-income	1,677	0,59629127

Logistic regression results indicate that the education variable is significant at the 1% level and positively correlated with the livelihood diversification (C = 2.999, Pr > χ^2 = 0.005). This suggests that a household head with a high level of education is more likely to diversify its livelihood. The high educational level of the household heads makes them able to look for employment activities and to undertake highly remunerative activities. This result is in line with previous researches, which found that the education is essential as it offers abilities and skills that enable households to secure well-paying jobs (Adebayo et al., 2012; Chaplin et al., 2002; Fabusoro et al., 2010). Thus, the vulnerability of their livelihoods may be reduced.
The findings from the logistic regression show that the farm size is significant at 5% threshold but with a negative sign (C =-2.364; Pr > χ^2= 0.014). This indicates that households with small farm size are more likely to diversify their livelihood strategies. The inverse relationship between the farm size and livelihood diversification have been revealed in several previous studies (Fabusoro et al., 2010; Adebayo et al., 2012; Sallawu et al., 2016). This affirmed that when agricultural activities offer sufficient revenue, households tend to decrease the diversification of their livelihood strategies. Indeed, households with large farm size are more likely to involve in agricultural activities as it requires more labor and enables them to ensure food security. Further, results in table 2 reported a positive coefficient of herd size, which was significant at 10% threshold. This indicates that the herd size was a determinant of household's livelihood diversification. Comparison of this finding with those of other studies confirms that the livestock endowment appears to be a crucial asset for improving diversification (Eneyew, 2012, Saha et al., 2015). A possible explanation for this might be that livestock help to generate more income and reduce liquidity constraints through selling animals. Moreover, the findings of this research suggest a significant (at 10% level) and a positive relationship of the non-farm income with livelihood diversification. The non-farm income is a source of investment for farm activities particularly in case of financial problem. It can further support the household's livelihood diversification and raise their standard of living. Finally, our model showed a positive and significant effect of the variable "distance to

Downloaded from https://cabidigitallibrary.org by 196.203.128.101, on 04/23/26.
Subject to the CABI Digital Library Terms & Conditions, available at https://cabidigitallibrary.org/terms-and-conditions

the main plot". We can conclude that the proximity to the main plot is one of the factors that facilitate the management of the farm.

Conclusion

The study analyzed the livelihood diversification of farm households in the province of Medenine-Tunisia. The diversification seems to be a common practice. It was infrequent to find a household without non-farm income. However, the level of diversification differs between households which affect their capital asset. It was found that households who highly diversify their livelihoods have more abundant human, social and financial capital. Many determinants either negatively or positively affected the livelihood diversification. On the first hand, the education, the herd size, the non-farm-income and the distance to the main plot have positive and significant effect on livelihood diversification. On the other hand, the farm size and the experience in the agricultural sector have significant but negative effect on livelihood diversification.

References

Adebayo, C. O., Akogwu, G. O., Yisa, E. S. (2012). Determinants of income diversification among farm households in Kaduna State: Application of Tobit regression model. Prod AgricTechnol J, 8(2), 1-10.

Ahmed, M. T., Bhandari, H., Gordoncillo, P. U., Quicoy, C. B., Carnaje, G. P. (2018). Factors affecting extent of rural livelihood diversification in selected areas of Bangladesh. SAARC Journal of Agriculture, 16(1), 7-21.

Asfaw, A., Simane, B., Hassen, A., & Bantider, A. (2017). Determinants of non-farm livelihood diversification: evidence from rainfed-dependent smallholder farmers in northcentral Ethiopia (Woleka sub-basin). Development Studies Research, 4(1), 22-36.

Babatunde, R. O., Qaim, M. (2009). Patterns of income diversification in rural Nigeria: determinants and impacts. Quarterly Journal of International Agriculture, 48(4), 305.

Carney, D. (2003). Sustainable livelihoods approaches: progress and possibilities for change (p.64). London: Department for International Development.

Chambers, R. (1987). Sustainable livelihoods, environment and development: putting poor rural people first.

Chaplin, H., Davidova, S., Gorton, M. (2002). Non-agricultural diversification of farm households and corporate farms in Central Europe (No. 1226-2016-98669).

Ellis, F. (1999). Rural livelihood diversity in developing countries: evidence and policy implications (Vol. 40, No. 1, pp. 1-10). London: Overseas Development Institute.

Fabusoro, E., Omotayo, A. M., Apantaku, S. O., Okuneye, P. A. (2010). Forms and determinants of rural livelihoods diversification in Ogun State, Nigeria. Journal of Sustainable Agriculture, 34(4), 417-438.

Gecho, Y. (2017). Rural farm households' income diversification: The case of Wolaita Zone, Southern Ethiopia. Social Sciences, 6(2), 45-56.

Karl, M. (2002). Participatory Policy Reform from a Sustainable Livelihoods Perspective. Food and Agriculture Organisation, Rome.

Khatun, D., & Roy, B. C. (2012). Rural livelihood diversification in West Bengal: determinants and constraints. Agricultural Economics Research Review, 25(347-2016-16910), 115.

McFadden, D. (1973). Conditional logit analysis of qualitative choice behavior.

Sallawu, H., Tanko, L., Nmadu, J. N., Ndanitsa, A. M. (2016). Determinants of income Diversification among farm households in niger State, Nigeria. Russian Journal of Agricultural and Socio-Economic Sciences, 50(2).

Downloaded from https://cabidigitallibrary.org by 196.203.128.101, on 04/23/26.
Subject to the CABI Digital Library Terms & Conditions, available at https://cabidigitallibrary.org/terms-and-conditions

Saha, B., & Bahal, R. (2015). Factors leading to success in diversified occupation: A Livelihood analysis in India. The Journal of Agricultural Education and Extension, 21(3), 249-266.

Xu, D., Zhang, J., Rasul, G., Liu, S., Xie, F., Cao, M., & Liu, E. (2015). Household livelihood strategies and dependence on agriculture in the mountainous settlements in the Three Gorges Reservoir Area, China. Sustainability, 7(5), 4850-4869.

Downloaded from https://cabidigitallibrary.org by 196.203.128.101, on 04/23/26.
Subject to the CABI Digital Library Terms & Conditions, available at https://cabidigitallibrary.org/terms-and-conditions